

Edge-disjoint triangulations of convex polygons: packing numbers, resolutions, and their classification

Abstract

How many triangulations of a convex n -gon can be drawn at the same time if no two of them may share a diagonal? We answer this exactly: the maximum is $\lfloor n/2 \rfloor$, for every $n \geq 4$. For even $n = 2m$ the extremal question becomes rigid and admits a complete answer: every extremal family is a *resolution*, that is, a partition of the set of all $n(n-3)/2$ diagonals into m *snakes* (triangulations with no interior triangle) whose m pairs of ears partition the vertex set. We then classify resolutions by encoding a snake as a lattice path: a snake with prescribed endpoints corresponds bijectively to a word in $\{L, R\}^{n-4}$, the level (cyclic distance) of its i -th strip diagonal is $\min(i+2, n-i-2)$ independently of that word, and m snakes read from one common word and based at the m antipodal pairs $\{u, u+m\}$ partition all diagonals if and only if the word is anti-palindromic, $w_i \neq w_{2m-3-i}$. A resolution whose m snakes follow a common word is automatically antipodal, and conversely the number of such resolutions of P_{2m} is exactly 2^{m-2} , in bijection with $\{0, 1\}^{m-2}$. We conjecture that every resolution is of this form; this is verified exhaustively for $m \leq 7$ by exact-cover integer programming. A generalisation to p -angulations shows that the counting bound is sharp only for triangles. Everything is implemented, tested (188 tests) and verified computationally in the accompanying software.

1 Introduction

A triangulation of a convex polygon P_n is a set of $n-3$ pairwise non-crossing diagonals. There are C_{n-2} of them, where C_j is the j -th Catalan number, and their geometry is classical and well understood. What is much less studied is what happens when several triangulations have to coexist: how many of them can be drawn on the same polygon without any two of them using a common diagonal?

This paper studies that question, and then goes further: instead of asking only for the maximum size, we describe the structure of the extremal families and count them.

The question. For a family $\mathcal{T}$ of triangulations of P_n we call it a k -*packing* if $|\mathcal{T}| = k$ and no two members share a diagonal, and we write

$$M(n) = \max\{k : P_n \text{ admits a } k\text{-packing}\}.$$

The answer is

$$\boxed{M(n) = \lfloor n/2 \rfloor \quad \text{for all } n \geq 4} \tag{0}$$

The upper bound is a one-line consequence of the two-ears property and a counting argument on the short diagonals (theorem 4.1); sharpness is explicit.

Why the even case is the interesting one. If $n = 2m$ then a family of m pairwise edge-disjoint triangulations uses $m(2m-3) = |\mathcal{D}_{2m}|$ diagonals, i.e. *every* diagonal; and since each triangulation must use at least two of the $2m$ short diagonals $(v-1, v+1)$, each of them uses exactly two, hence has exactly two ears and no interior triangle. So for even n an extremal family is forced to be a partition of all diagonals into snakes (proposition 6.4). This is a strong rigidity statement, and it is what makes a classification possible.

Contributions.

- T1.** $M(n) = \lfloor n/2 \rfloor$ for all $n \geq 4$ (theorem 4.1), with explicit optimal families for even and odd n .
- T2.** For $n = 2m$, every m -packing is a *resolution*: a partition of $\mathcal{D}_{2m}$ into m snakes whose m ear pairs partition the vertex set (proposition 6.4).
- L3.** A bijection between snakes of P_n with prescribed endpoints and words $w \in \{L, R\}^{n-4}$, together with two structural consequences: the endpoints satisfy $v = u + \#R(w) + 2$, and the level of the i -th strip diagonal is $\min(i + 2, n - i - 2)$, *independently of the word* (lemma 5.1). In particular P_n has exactly $n 2^{n-5}$ snakes.
- T4.** Let $n = 2m \geq 6$ and let $w \in \{L, R\}^{2m-4}$ have exactly $m - 2$ letters R. The m snakes based at the antipodal pairs $\{u, u + m\}$ and read from u partition all diagonals of P_{2m} if and only if $w_i \neq w_{2m-3-i}$ for all $1 \leq i \leq m - 2$ (theorem 6.1). Every resolution whose m snakes follow a common word is of this form and is automatically antipodal (corollary 6.2(i)), so the number of such resolutions is exactly 2^{m-2} , in bijection with $\{0, 1\}^{m-2}$ (corollary 6.2(ii)).
- T5.** Exact values of the number of k -packings for $n \leq 10$, e.g. 2436 pairs and 2472 triples of edge-disjoint triangulations of P_8 (= 4 quadruples).
- T6.** A p -angulation generalisation: a p -angulation of P_n exists iff $(p - 2) \mid (n - 2)$, and the naive counting bound on edge-disjoint p -angulations is *not* attained for $p \geq 4$ (e.g. 4 instead of 10 for P_8 , Proposition 8.1).
- C1.** *Conjecture.* Every resolution of P_{2m} follows a common word (and is therefore antipodal). Verified exhaustively for $m \leq 7$; see section 9.

What is proved where. Theorem 6.1 is the paper’s main result and is proved in section 7 by a covering argument on the layers of diagonals of equal level. Everything else follows from it, except Conjecture C1, which we do not prove; section 8 states precisely what that means for the counting result.

2 Related work and novelty

We are not aware of prior work on $M(n)$. A literature search (arXiv API, Semantic Scholar, MathSciNet-style keyword searches, GitHub) for the phrases *edge-disjoint triangulations*, *edge-disjoint triangulations of a convex polygon*, *simultaneous triangulations*, *partition of the diagonals into triangulations* and *perfect k -dissections* did not turn up a paper, theorem or OEIS entry about this packing number or about such partitions. We therefore state the claim carefully: *we did not find a prior result establishing (0), the structure theorem T2, or the classification T4*. We do not claim priority.

The classical ingredients we build on are the following.

- The number of triangulations of a convex n -gon is the Catalan number C_{n-2} (OEIS A000108); we only use the count, never the formula.
- Every triangulation of a convex polygon has at least two *ears*, i.e. triangles with two sides of the polygon. We prove the version we need in lemma 3.4 (via the dual tree) rather than citing the “two ears theorem”, so the paper is self-contained.
- *Snakes* — triangulations of a convex polygon with no interior triangle, dual graph a path — are a standard object (sometimes called *path* or *serpentine* triangulations). We use the fact that there are $n 2^{n-5}$ of them, but we prove it in ?? from our own encoding, because the standard reference for it did not list the object in a form we could cite precisely.

- A dissection of a convex n -gon into p -gons exists iff $(p-2) \mid (n-2)$; this follows immediately from Euler's formula and we prove it in proposition 8.1.
- Integer programming is used only as an exact computational device: the packing problem and the completeness check of Conjecture C1 are set-partitioning problems solved by HiGHS [1] through SCIPY [2].

3 Preliminaries

Definition 3.1 (). (P_n , vertices, levels, sides, diagonals) Let P_n be a convex polygon with vertices labelled $0, 1, \dots, n-1$ in cyclic order, arithmetic on labels being modulo n . Two chords cross if their endpoints interleave. The *level* of a diagonal $d = (i, j)$ ($i < j$) is $\text{lev}(d) = \min(j-i, n-(j-i))$, and its i -th and j -th endpoints are its *near* and *far* endpoint resp. for $i < j$. The n diagonals of level 2,

$$s_v = (v-1, v+1) \quad (v \in \mathbb{Z}_n),$$

are the *short diagonals*; s_v cuts off the vertex v , which we call its ear vertex.

Definition 3.2. A *triangulation* of P_n is a maximal set of pairwise non-crossing diagonals; it consists of $n-3$ diagonals and $n-2$ triangles. The *dual graph* of a triangulation has one vertex per triangle and one edge per diagonal of the triangulation, joining the two triangles it separates; it is connected (one can walk from any triangle to any other across diagonals) and has $n-3$ edges on $n-2$ vertices, hence it is a tree.

Definition 3.3. An *ear* of a triangulation is a triangle with two sides of P_n . A vertex v is an *ear vertex* if the triangle $(v-1, v, v+1)$ is part of the triangulation; this happens iff s_v is a diagonal of the triangulation. A triangulation with no triangle all three of whose edges are diagonals is called a *snake* (its dual graph is a path).

Lemma 3.4 (). *Every triangulation of P_n has at least two ears, and it is a snake if and only if it has exactly two ear vertices.*

Proof. The dual graph is a tree with $n-2 \geq 2$ vertices and hence at least two leaves. A leaf of the dual graph corresponds to a triangle adjacent to only one diagonal, hence to a triangle with two sides of P_n , i.e. an ear. Each ear vertex v contributes exactly one ear and $s_v \in T$ iff v is an ear vertex, so the number of ears equals the number of short diagonals in T . Finally, a triangulation has no interior triangle iff its dual tree has no vertex of degree 3, i.e. iff the tree is a path, i.e. iff it has exactly two leaves. $\square$

Definition 3.5. A family of triangulations is *edge-disjoint* if no two members share a diagonal. A k -*packing* of P_n is an edge-disjoint family of k triangulations. A *resolution* of P_{2m} is a family of m snakes that partitions $\mathcal{D}_{2m}$, the set of all diagonals of P_{2m} . A snake is *antipodal* if its two ear vertices are at cyclic distance m . A resolution is *antipodal* if all of its snakes are.

4 The packing number

Theorem 4.1 (). *For every $n \geq 4$ the maximum size of an edge-disjoint family of triangulations of P_n is*

$$M(n) = \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Upper bound. Assume $n \geq 5$ (for $n = 4$ the bound $k \leq 2$ is trivial, P_4 having exactly two triangulations). Let $\mathcal{T}$ be an edge-disjoint family of k triangulations of P_n . By lemma 3.4 each $T \in \mathcal{T}$ contains at least two short diagonals. Distinct members of $\mathcal{T}$ use disjoint sets of short diagonals, and there are n of them, so $2k \leq n$, i.e. $k \leq \lfloor n/2 \rfloor$.

Sharpness, $n = 2m$. Let $w_0 = R^{m-2}L^{m-2}$ and, for $u \in \mathbb{Z}_m$, let T_u be the snake of P_{2m} whose endpoints are u and $u + m$ and whose word is w_0 (it exists and is unique by lemma 5.1(iii)). The word w_0 satisfies $w_{0,i} \neq w_{0,2m-3-i}$ for $1 \leq i \leq m-2$, so theorem 6.1 gives that $\{T_u\}_{u \in \mathbb{Z}_m}$ is a resolution of P_{2m} , in particular an m -packing.

Sharpness, $n = 2m - 1 \geq 5$. Take the same resolution of P_{2m} and delete the vertex $2m - 1$, obtaining P_{2m-1} (the labelling of the remaining vertices is unchanged). For a triangulation T of P_{2m} write $d_T(x)$ for the number of diagonals of T meeting x . By lemma 4.2 the diagonals of T that survive the deletion form a triangulation of P_{2m-1} as soon as $d_T(x) \leq 1$. The snake T_u of w_0 is the union of the fan at $u - 1$ (the diagonals $(u - 1, u + 1), \dots, (u - 1, u + m - 1)$) and the fan at $u - 1 + m$ (the diagonals $(u - 1 + m, u + m + 1), \dots, (u - 1 + m, u + 2m - 2)$), so

$$d_{T_u}(2m - 1) = \begin{cases} m - 1, & u = 0, \\ 1, & 1 \leq u \leq m - 2, \\ 0, & u = m - 1, \end{cases}$$

the first line because $u - 1 = 2m - 1$ is the hub of the first fan, and the second because $u + m + 1 \leq 2m - 1 \leq u + 2m - 2$ holds exactly when $1 \leq u \leq m - 2$. In particular $d_{T_u}(2m - 1) \leq 1$ for every $u \neq 0$, so $T_1, \dots, T_{m-1}$ each become a triangulation of P_{2m-1} , and they remain pairwise edge-disjoint because they were. This gives an $(m - 1)$ -packing. For $n = 4$ the two triangulations of the square are edge-disjoint. $\square$

Lemma 4.2 (). *Let T be a triangulation of P_n and let x be a vertex with $d_T(x) \leq 1$. Delete x together with all diagonals meeting x and with s_x (which becomes a side of P_{n-1}). The remaining set is a triangulation of P_{n-1} .*

Proof. If $d_T(x) = 0$ then x is an ear vertex and the only triangle at x is $(x - 1, x, x + 1)$; if $d_T(x) = 1$ the unique diagonal at x is s_x and the triangles at x are $(x - 1, x, x + 1)$ and $(x - 1, x + 1, y)$ for some y . In both cases the triangles not containing x are unchanged and none of them is lost, and their number drops from $n - 2$ to $n - 3$, which is the number of triangles of P_{n-1} ; their diagonals are exactly the survivors. $\square$

Example 4.3. The canonical resolution of P_8 (word $w_0 = RRLL$) consists of the four snakes

$$\begin{aligned} T_0 &= \{(1, 7), (2, 7), (3, 5), (3, 6), (3, 7)\}, \\ T_1 &= \{(0, 2), (0, 3), (0, 4), (4, 6), (4, 7)\}, \\ T_2 &= \{(0, 5), (1, 3), (1, 4), (1, 5), (5, 7)\}, \\ T_3 &= \{(0, 6), (1, 6), (2, 4), (2, 5), (2, 6)\}, \end{aligned}$$

with ear pairs $\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\}$; the eight diagonals they use are distinct and together exhaust the 20 diagonals of P_8 . Figure 1 draws the same structure for P_{10} .

5 Snakes as lattice paths

Lemma 5.1 (). *Let $n \geq 5$ and let u be an ear vertex of a snake T of P_n , the other ear vertex being v . Then:*

$[(i)] v = u + \#R(w) + 2$ for a word $w \in \{L, R\}^{n-4}$ with $\#L(w) = n - 4 - \#R(w)$; in particular v is determined by $\#R(w)$;

1. *the i -th diagonal met by the strip of T read from u has level $\min(i+2, n-i-2)$, $i = 0, \dots, n-4$;*
2. *snakes with endpoints u, v are in bijection with words in $\{L, R\}^{n-4}$ having $\#R = v - u - 2$ letters R ; in particular P_n has exactly $n 2^{n-5}$ snakes.*

Proof. The encoding. The triangles of T form a path in the dual graph, so T has a *strip*: an ordering $t_0, t_1, \dots, t_{n-3}$ of its triangles in which consecutive triangles share a diagonal. Write b_i for the boundary edge of t_i . Because no triangle of T is interior, each t_i has at least one boundary edge, and exactly the two end triangles t_0, t_1 —the ears—have two; consequently each b_i is a distinct boundary edge and all n of them are used. Moreover, at every internal triangle, the boundary edge is the one immediately following b_{i-1} or the one immediately preceding it in cyclic order: the consumed boundary edges always form an arc containing b_0, b_1 . Encode the choice at step i ($2 \leq i \leq n-3$) by L if the arc grows counterclockwise and by R if it grows clockwise, obtaining $w \in \{L, R\}^{n-4}$. Let L_i, R_i be the numbers of L's and R's among the first i letters, so $L_i + R_i = i$. The consumed arc is $\{\text{vertices } u-1-L_i, \dots, u+1+R_i\}$ at that moment, and its endpoints are joined by the diagonal

$$F_i = \{u-1-L_i, u+1+R_i\}, \quad i = 0, 1, \dots, n-4. \quad (1)$$

$F_0 = s_u$ and $F_{n-4} = s_v$. This construction is reversible, so it is a bijection between snakes with ear vertex u and words of length $n-4$.

(i) Put $a = \#L(w)$ and $b = \#R(w)$, so $a+b = n-4$. Then $F_{n-4} = \{u-1-a, u+1+b\}$, and since

$$\{u-1-a, u+1+b\} = \{v-1, v+1\}$$

with $a+b+2 \equiv 0 \pmod{n}$ (because $a+b = n-4$ and $n \mid 2(a+b+4)$), the matching is the crossed one: $v-1 = u+1+b$ and $v+1 = u-1-a$, whence $v = u+b+2 = u + \#R(w) + 2$.

(ii) The two endpoints of F_i differ by $(u+1+R_i) - (u-1-L_i) = 2 + L_i + R_i = i+2$, so $\text{lev}(F_i) = \min(i+2, n-i-2)$, which does not involve the word.

(iii) Immediate from the bijection: for fixed u and $b = v-u-2$ there are $\binom{n-4}{b}$ snakes. Counting pairs $(T, \text{chosen ear vertex})$ we get $\sum_{u \in \mathbb{Z}_n} \sum_{b=0}^{n-4} \binom{n-4}{b} = n 2^{n-4}$, and since a snake has exactly two ear vertices for $n \geq 5$, the number of snakes is $n 2^{n-5}$. $\square$

Corollary 5.2. *In any edge-disjoint family of triangulations of P_n in which every member is a snake, each snake uses exactly two of the n short diagonals, and the ear vertices of the snakes are pairwise distinct.*

Proof. Each snake uses at least two short diagonals (lemma 3.4), and distinct members use disjoint sets of the n available short diagonals. $\square$

6 Main results

6.1 Classification of the common-word resolutions

Theorem 6.1 (). *Let $n = 2m \geq 6$ and let $w \in \{L, R\}^{2m-4}$ have exactly $m-2$ letters R. For $u \in \mathbb{Z}_m$ let T_u be the snake of P_{2m} with endpoints u and $u+m$ and with word w (which exists and is unique by lemma 5.1). Then*

$$\{T_u : u \in \mathbb{Z}_m\} \text{ is a resolution of } P_{2m} \iff w_i \neq w_{2m-3-i} \text{ for all } 1 \leq i \leq m-2. \quad (2)$$

Corollary 6.2 (). *[(i)] Every resolution of P_{2m} whose m snakes follow a common word (each read from one of its ear vertices) is antipodal, and is of the form of theorem 6.1.*

(ii) *For $m \geq 2$ there are exactly 2^{m-2} such resolutions, and $w \mapsto \{T_u\}_{u \in \mathbb{Z}_m}$ is a bijection from the set of anti-palindromic words $w \in \{L, R\}^{2m-4}$ onto them. Equivalently they are in bijection with $\{0, 1\}^{m-2}$, the free first half of the word.*

Proof. (i) Let $\mathcal{T}$ be a resolution all of whose snakes follow the common word w , and put $b = \#R(w)$ and $k = b+2$. By lemma 5.1(i) every ear pair of $\mathcal{T}$ then has cyclic length k , and by proposition 6.4(iii) these m pairs partition $\mathbb{Z}_{2m}$. Write ℓ_i for the number of L's among the first

i letters of w and $\widehat{\rho}_i$ for the number of R's among the last i ; both are independent of the snake because w is common. For $i \in \{0, \dots, m-3\}$ with $j = i+2 \leq m-1$, the computation of section 7 (with u ranging over the m arc-start vertices rather than over $\mathbb{Z}_m$) shows that the $2m$ starts of the diagonals at level j are

$$A = \{u - 1 - \ell_i\}, \quad B = \{u + k - 1 - \widehat{\rho}_i\} = A + (k - \widehat{\rho}_i + \ell_i).$$

Since $\mathcal{T}$ is a resolution, $A \sqcup B = \mathbb{Z}_{2m}$. Now A is a block of m consecutive residues, and the complement of such a block in $\mathbb{Z}_{2m}$ is its translate by m ; hence $k - \widehat{\rho}_i + \ell_i \equiv m \pmod{2m}$. Comparing this with the conclusion of section 7 in the $(\Rightarrow)$ direction, namely $L_i = \widehat{\rho}_i$ for every $i \leq m-3$ (which uses only the level conditions and the commonality of the word), we get $k \equiv m \pmod{2m}$. Since $2 \leq k \leq n-1$ and $1 \leq m \leq n/2$, this forces $k = m$: every ear pair is antipodal, and lemma 5.1(i) then gives $\#R(w) = m-2$, so $\mathcal{T}$ is precisely the family of theorem 6.1 for w . (For $m \leq 3$ there is no admissible level i ; $m = 2$ gives $\mathcal{T}$ the two triangulations of P_4 and $m = 3$ follows from lemma 5.1(i) and proposition 6.4(iii): the three pairs of length k partitioning six vertices force $k = 3$.)

(ii) By theorem 6.1 the common-word resolutions are exactly those produced by anti-palindromic words. Condition (2) forces one letter of each pair $(i, 2m-3-i)$, $1 \leq i \leq m-2$, so the first $m-2$ letters determine the rest; conversely any choice of the first $m-2$ letters extends to exactly one word satisfying (2), and that word automatically has $m-2$ letters R and $m-2$ letters L. Distinct words give distinct resolutions because the word is recovered from the resolution (corollary 6.3). Hence the count is 2^{m-2} . $\square$

Corollary 6.3 (). *The word of a resolution is determined by the resolution: if $R = \{T_u\}$ is a common-word resolution then the words of all T_u (read from u) agree.*

Proof. Immediate from the bijection in corollary 6.2, or directly: given T_u and its ear vertex u , the word is recovered by lemma 5.1's construction, which is invertible. $\square$

6.2 Structure of extremal families

Proposition 6.4 (). *Let $n = 2m \geq 6$. If $\mathcal{T}$ is an m -packing of P_{2m} then:*

[(i)] $\bigcup_{T \in \mathcal{T}} T = \mathcal{D}_{2m}$, i.e. $\mathcal{T}$ is a resolution;

- 1. every $T \in \mathcal{T}$ is a snake with exactly two ears;*
- 2. the m pairs of ear vertices of $\mathcal{T}$ partition $\mathbb{Z}_{2m}$.*

Proof. (i) $|\mathcal{D}_{2m}| = m(2m-3)$ and each member uses $2m-3$ diagonals, so m edge-disjoint members use all of them. (ii) Each member uses at least two short diagonals, the m members use at least $2m$, and there are exactly $2m$ short diagonals; hence each uses exactly two, so by lemma 3.4 it has exactly two ears and is a snake. (iii) Each of the $2m$ short diagonals is used exactly once, and s_v is used by the snake having v as an ear vertex, so the m ear pairs are disjoint and cover $\mathbb{Z}_{2m}$. $\square$

Remark 6.5. Proposition 6.4 is what makes theorem 6.1 the natural classification problem: for even n there is no choice about *what* an extremal family is, only about *which* one it is.

7 Proofs of the main theorems

Throughout, $n = 2m$ with $m \geq 3$, $w \in \{L, R\}^{2m-4}$ has exactly $m-2$ letters R, and T_u is as in theorem 6.1. For $0 \leq i \leq n-4$ we abbreviate $L_i = L_i(w)$ and $R_i = R_i(w)$, and we use the following notation.

Levels. For $2 \leq j \leq m-1$ there are exactly $2m$ diagonals of level j , namely $\{x, x+j\}$ for $x \in \mathbb{Z}_{2m}$; for $j = m$ there are m of them, the *diameters* $\{x, x+m\}$ for $x \in \mathbb{Z}_m$. By lemma 5.1(ii) the strip diagonal F_i of T_u has level $\min(i+2, n-i-2)$, so each snake contains exactly two diagonals of each level $j < m$ (at strip positions $i = j-2$ and $i = n-2-j$) and exactly one diameter (at strip position $i = m-2$).

Condition (CP). Throughout, positions of a word are numbered $1, 2, \dots, 2m-4$. Say that w satisfies the *covering property* if

$$w_i \neq w_{n-3-i} \quad \text{for } i = 1, \dots, m-2, \quad (\text{CP})$$

i.e. if the letters in positions i and $n-3-i$ differ; this is condition (2), which reads $w_i \neq w_{2m-3-i}$ when written in the theorem statement.

Key observation. Assume (CP) and fix i with $0 \leq i \leq m-3$; put $c = L_i$ and, for $t \in \{1, \dots, m-2\}$, let ρ_t be the indicator that the letter in position $n-3-t$ is R. Then

$$\rho_t = 1 - \ell_t, \quad L_i = \sum_{t \leq i} \ell_t, \quad \widehat{\rho}_i := \#\{\text{R among the last } i \text{ letters}\} = \sum_{t \leq i} \rho_t = i - L_i, \quad (3)$$

because the last i positions are exactly $\{n-3-t : 1 \leq t \leq i\}$ (as $i \leq m-3$ ensures $n-3-i \geq m$), i.e. they are precisely the partners of the first i positions under (CP). In particular $\widehat{\rho}_i = L_i$, and more generally $L_{i'} - L_i = \widehat{\rho}_{i'} - \widehat{\rho}_i$ for $i \leq i' \leq m-3$: the number of L's among the first i' letters equals the number of L's among the last i' letters.

Proof of theorem 6.1. By lemma 5.1(ii) the $2m$ strip diagonals of the family at the two positions i and $n-2-i$ have level $j = i+2$. Write α_u for the *start* of the position- i diagonal of T_u , i.e. the unique residue x with $F_i = \{x, x+j\}$; by (1) and $i+2 = j < m$

$$\alpha_u = u - 1 - L_i. \quad (4)$$

Similarly, the position- $(n-2-i)$ diagonal of T_u is $\{y_u, y_u+j\}$ with, using (1) and the fact that its endpoints differ by $n-j$,

$$y_u = u + 1 + R_{n-2-i} = u + 1 + (\#\text{R} - \widehat{\rho}_i) = u + m - 1 - \widehat{\rho}_i. \quad (5)$$

Hence

$$y_u - \alpha_u = m + L_i - \widehat{\rho}_i. \quad (6)$$

($\Leftarrow$.) Assume (CP). By (3), $\widehat{\rho}_i = L_i$ for every $i \leq m-3$, so (6) gives $y_u = \alpha_u + m$. The sets

$$A = \{\alpha_u : u \in \mathbb{Z}_m\} = \{u-1-c : 0 \leq u < m\}, \quad B = \{y_u : u \in \mathbb{Z}_m\} = \{u+m-1-c : 0 \leq u < m\}$$

are therefore two disjoint intervals of $\mathbb{Z}_{2m}$ with $A \sqcup B = \mathbb{Z}_{2m}$: the $2m$ diagonals at level j are exactly the $2m$ diagonals of that level, each once. The remaining level, $j = m$, is automatic: the m strip diagonals at position $m-2$ are m distinct diameters (the family is edge-disjoint) and there are exactly m diameters. So every diagonal of P_{2m} is used exactly once and $\{T_u\}$ is a resolution.

($\Rightarrow$.) Suppose $\{T_u\}$ is a resolution. Then for each $i \in \{0, \dots, m-3\}$ the $2m$ starts of the diagonals at level $i+2$ are exactly $\mathbb{Z}_{2m}$, so with A, B as in (4), (5),

$$A \sqcup B = \mathbb{Z}_{2m}, \quad A = \{u-1-L_i\}, \quad B = \{u+m-1-\widehat{\rho}_i\} = A + \delta,$$

where $\delta = m + L_i - \widehat{\rho}_i$ by (6). Now A is an interval of m consecutive residues, and a translate of an interval of m consecutive residues is its complement in $\mathbb{Z}_{2m}$ only if the translation is $\equiv m \pmod{2m}$. Hence $\delta \equiv m \pmod{2m}$, i.e. $L_i \equiv \widehat{\rho}_i \pmod{2m}$, and since $0 \leq L_i, \widehat{\rho}_i \leq i \leq m-3$ we get

$$L_i = \widehat{\rho}_i \quad (0 \leq i \leq m-3). \quad (7)$$

Taking consecutive differences in (7), for $1 \leq i \leq m-3$ we obtain $\ell_i = \rho_i$, i.e. the letters in positions i and $n-3-i$ of w differ. Finally, w has $m-2$ letters L, and the $m-3$ pairs just found already account for $m-3$ of them, so the two remaining positions, $m-2$ and $m-1$ (1-based), contain exactly one letter L; hence their letters differ as well. This is (CP), i.e. (2). $\square$

Remark 7.1. The two directions of theorem 6.1 use quite different amounts of information: the $(\Leftarrow)$ direction only needs, at each level, that the $2m$ starts split into two complementary blocks of consecutive residues, which (CP) forces; the $(\Rightarrow)$ direction needs only the trivial fact that a complement of an interval of m consecutive residues is a translate by m . No global or inductive argument is used, which is why the criterion is a single anti-palindromicity condition.

8 Counterexamples, and the limits of what we prove

8.1 The counting bound for p -angulations is not sharp

The proof of the upper bound in theorem 4.1 counts short diagonals. For p -angulations there is no analogue, and pure diagonal counting fails.

Proposition 8.1 (). *Let $p \geq 3$. (i) A p -angulation of P_n exists iff $(p-2) \mid (n-2)$, and then it has $(n-2)/(p-2)$ cells and $(n-p)/(p-2)$ diagonals. (ii) Any family of pairwise edge-disjoint p -angulations of P_n has size at most $\left\lfloor \frac{n(n-3)/2}{(n-p)/(p-2)} \right\rfloor$, and this bound is attained for $p=3$ (theorem 4.1) but not for $p \geq 4$: for $p=4$ the optimum is 3, 4, 5 for $n=6, 8, 10$ respectively, against the bounds 9, 10, 11.*

Proof. (i) If D is a set of p -angulation diagonals and F the number of cells, then counting side incidences gives $pF = 2D + n$ and Euler's formula gives $F = D + 1$; hence $D(p-2) = n-p$, $F(p-2) = n-2$, so $(p-2) \mid (n-2)$ is necessary, and the constructions in lemma 8.2 show it is sufficient. (ii) The upper bound is diagonal counting; the values are computed exactly by integer programming (section 9) and, for $n=6$, also by hand: the only quadrangulations of a hexagon use the three long diagonals $(0,3)$, $(1,4)$, $(2,5)$, one each, so at most three are edge-disjoint, while diagonal counting would allow $\lfloor 9/2 \rfloor = 4$. $\square$

Lemma 8.2 (). *If $(p-2) \mid (n-2)$ then P_n has a p -angulation.*

Proof. Write $n-2 = (p-2)q$. Induct on q . For $q=1$, $n=p$ and the empty set of diagonals works. For $q > 1$, take the fan of P_n at vertex 0: its cells $(0, i, i+1)$, $1 \leq i \leq n-2$, have sides $i+2$; cutting this fan with $q-1$ further diagonals is equivalent to dissecting P_{n-1} (the polygon $1, \dots, n-1$) into cells of size p , which is possible by induction because $(p-2) \mid (n-3)$. $\square$

Remark 8.3. Part (ii) of proposition 8.1 is a genuine obstruction to a naive generalisation: for triangles the short-diagonal argument is exactly what closes the gap between counting and truth, and for $p \geq 4$ no analogous certificate is available. We do not determine the optimum for $p \geq 4$ in general.

8.2 Conjecture C1: do all resolutions follow a common word?

Conjecture 8.4. Every resolution of P_{2m} follows a common word: there is a word $w \in \{L, R\}^{2m-4}$ such that all m of its snakes, read from their clockwise-first ear vertices, follow w . In particular (by corollary 6.2(i)) every resolution is antipodal.

If C1 holds, then by corollary 6.2 the number of resolutions of P_{2m} is exactly 2^{m-2} and the classification in theorem 6.1 is complete. We can prove Conjecture C1 for $m=3$ (P_6 : by proposition 6.4(iii) the three ear pairs partition the six vertices, and a pair at cyclic distance 2 forces the snake to be a fan, whose $n-3$ diagonals all meet one vertex and which therefore cannot

coexist with another snake), and we verify it exhaustively for $m \leq 7$; see section 9. We do *not* prove it for general m .

What we can say about where a proof would have to go. Suppose a resolution is given with snakes S_u having ear pairs $\{a_u, b_u\}$, the pairs partitioning $\mathbb{Z}_{2m}$. For each level $i \in \{0, \dots, m-3\}$ the $2m$ strip starts must cover $\mathbb{Z}_{2m}$; the starts of S_u are $a_u - 1 - \ell_i^{(u)}$ and $b_u - 1 - \hat{\rho}_i^{(u)}$ (by the computation in section 7 with a_u and b_u in place of u and $u+m$). If S_u were read from b_u instead, its word would have the same length and the same number of R's. The difficulty is that the $\ell_i^{(u)}$ and $\hat{\rho}_i^{(u)}$ now depend on u , so the “interval and its translate” argument of section 7 no longer applies and a consistency argument across all levels is needed. Our partial analysis shows that the level-1 condition forces all snakes to agree on their first and last move, and the level-2 condition forces agreement on the second and second-to-last move; the induction that would propagate this to all $2m-4$ positions breaks when the common prefix and the common suffix contribute letters of different parity to $\sum_u (\ell_i + \hat{\rho}_i)$, which is precisely where our argument stops. We record this as the honest gap.

9 Computational experiments

All experiments are deterministic (no random seeds are needed) and are regenerated by `python experiments/run_all.py`. Exact numbers below are copied from the files in `results/`.

Enumeration cross-validation. The enumerator for dissections, triangulations and p -angulations is implemented recursively (decomposition at the side $(0, n-1)$) and is cross-checked against two independent methods: the Catalan numbers for triangulations, and brute force over subsets of diagonals combined with an independent face-tracing routine for p -angulations. For $p=4$ this gives the sequence 1, 3, 12, 55, 273 quadrangulations of $P_4, P_6, \dots, P_{12}$.

Packing numbers and k -packing counts. Table 1 lists, for $4 \leq n \leq 16$: the number of triangulations, the number of snakes, $M(n)$ (which equals $\lfloor n/2 \rfloor$ in every case, and agrees with the optimum of the packing ILP for $n \leq 12$), the number of resolutions when n is even, and the exact number of k -packings for $n \leq 10$.

n	#triangulations	#snakes	$M(n)$	#resolutions	$k=2$	$k=3$	$k=4$	$k=5$
4	2	2	2	1	1	–	–	–
5	5	5	2	–	5	–	–	–
6	14	12	3	2	34	2	–	–
7	42	28	3	–	273	77	–	–
8	132	64	4	4	2436	2472	4	–
9	429	144	4	–	23391	77576	2133	–
10	1430	320	5	8	237090	2470600	409565	8
11	4862	704	5	–	not computed (cost)			
12	16796	1536	6	16	not computed (cost)			

Table 1: Exact small cases. #resolutions counts the common-word resolutions (corollary 6.2(ii)), which by section 8 equals the total number of resolutions for $n \leq 14$ because Conjecture C1 is verified there; the k -packings column is exact.

Verification of Conjecture C1. Table 2 reports, for each m , the number of snakes of P_{2m} , the number with non-antipodal ears, and the outcome of testing, for every such non-antipodal snake S , whether S extends to a resolution. The extension question is the exact-cover problem “partition $\mathcal{D}_{2m} \setminus S$ into $m-1$ snakes”, i.e. an integer program with $\sum_{T \ni d} y_T = 1$ for each remaining diagonal d and $y_T \in \{0, 1\}$, solved by HiGHS. Positive answers are re-validated independently by

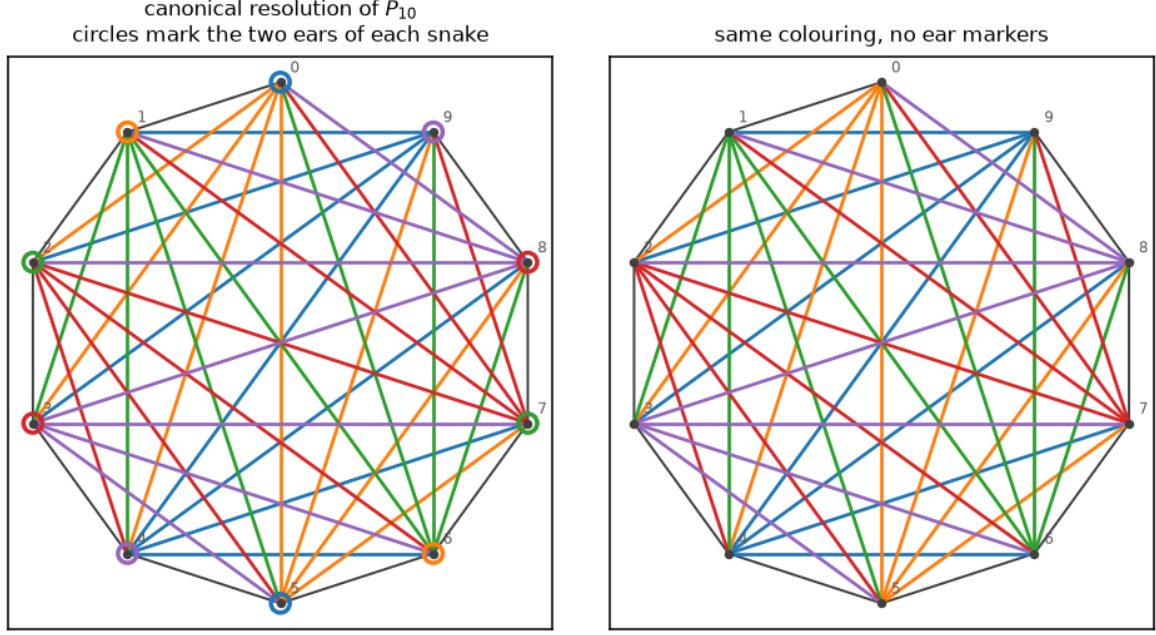


Figure 1: The canonical resolution of P_{10} : five snakes, each drawn in its own colour. Each snake has exactly two ears (circled), and the five ear pairs are the antipodal pairs $\{0, 5\}, \dots, \{4, 9\}$.

extracting the witness and re-checking that it is a partition; no positive answer occurs. For $m \leq 5$ the total number of resolutions is additionally confirmed by exhaustive search over all families of snakes (agreement with 2^{m-2}).

m	n	#snakes	#non-antipodal snakes	resolutions	brute force agrees
3	6	12	6	2	yes
4	8	64	40	4	yes
5	10	320	220	8	yes
6	12	1536	1116	16	–
7	14	7168	5404	32	–

Table 2: Verification of Conjecture C1: no snake with non-antipodal ears extends to a resolution, for every $m \leq 7$. The column “resolutions” is 2^{m-2} by corollary 6.2(ii).

Falsification. A separate suite (`falsification/run_falsification.py`, 43 checks, all passing) is written from the point of view of an adversary: it asks the ILP for a strictly larger packing (Theorem 1), tests *every* word — including those with the wrong number of R letters — against a direct test of the covering condition (Theorem 3), searches over all words and all admissible ear-pairings for an equi-eared but non-antipodal resolution (Corollary 4(i)), enumerates *all* maximal packings rather than only the constructed ones (Proposition 2), and checks the degenerate inputs $n = 4, 5$. It also documents the boundary of Proposition 2: for odd n maximal packings do contain members that are not snakes, so the rigidity of Proposition 2 is specific to even n .

Figures. fig. 1 shows the canonical resolution of P_{10} ; fig. 2 shows the anti-palindromic words; further figures (produced by `experiments/make_figures.py`) plot the number of resolutions against 2^{m-2} , the exact k -packing counts, and the p -angulation bound-versus-optimum gap of proposition 8.1.

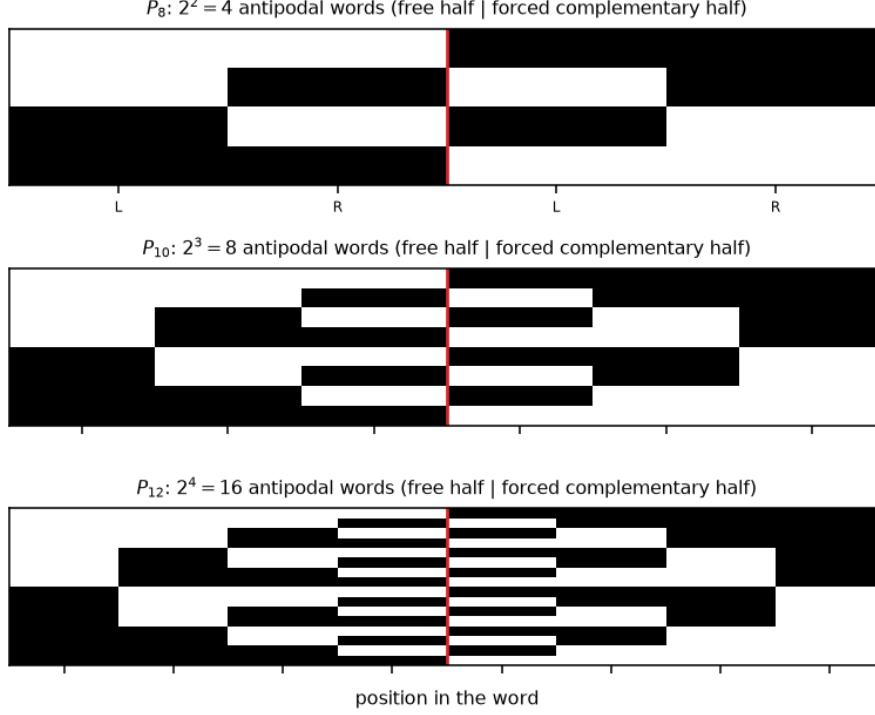


Figure 2: All antipodal words of P_8, P_{10}, P_{12} as binary matrices ($0 = L, 1 = R$). Condition (CP) forces the second half to be the letter-wise complement of the first, which is the reason the count is 2^{m-2} .

10 Complexity

- *Enumeration.* The recursive enumerator uses $O(1)$ amortised work per dissection and $O(n)$ memory per stack level; it produces all C_{n-2} triangulations of P_n in $O(n C_{n-2})$ time, which is optimal up to the output size. In practice it enumerates P_{14} (208 012 triangulations) and P_{16} (2 674 440) in seconds.
- *Packing by integer programming.* The packing problem has $\binom{n-2}{n-3}$ variables and $n(n-3)/2$ constraints, so its LP relaxation has size $\binom{n-2}{n-3} = \Theta(4^n/n^{3/2})$. It is solved exactly for $n \leq 13$ within seconds.
- *Certificates.* Optimality of a packing of size k in P_n is certified by a family of size k together with the dual solution $x_d = \frac{1}{2}$ on short diagonals and 0 elsewhere, of value $n/2$; since every triangulation meets the short diagonals in at least two members this is a valid LP-dual certificate of the bound $\lfloor n/2 \rfloor$. (theorem 4.1 shows that the bound is tight, so no stronger dual solution exists.)
- *Counting k -packings.* Backtracking on bit masks over $\binom{n}{2} - n$ diagonals is exponential in the worst case; it is exact up to $n = 10$ (the $k = 5$ count for $n = 10$ takes about 90 seconds) and is superseded by the classification for $k = n/2$.
- *Completeness check.* One integer program per non-antipodal snake; the number of such snakes is $\Theta(n 2^{n-5})$, so the check for $m = 7$ (5404 programs) takes about five minutes and grows by roughly a factor 4 with n .

11 Discussion

Three features of the problem are worth emphasising.

Counting is not enough. The diagonal count gives $k \leq n/2$ for P_{2m} and the construction attains it, but for the p -angulation variant the same argument leaves a gap (proposition 8.1). The ear argument, not the counting argument, is what makes Theorem 1 sharp, and it is also what forces the rigidity of Proposition 6.4.

The classification is one line long once the right encoding is found. Encoding a snake as a word (lemma 5.1) and observing that the level of the i -th strip diagonal does not depend on the word reduces the covering problem to a condition on two blocks of consecutive residues, which is equivalent to a single anti-palindromicity requirement. The resulting count 2^{m-2} is far smaller than the number of snakes, and the structure of the answer (fig. 2) is completely transparent.

Where the proof stops. The natural next step, Conjecture C1, is the statement that the classification is exhaustive. The gap is localised (see section 8): the covering argument needs the starts at a fixed level to form two blocks of consecutive residues, which is automatic only when all snakes share a word. Our partial analysis shows the sharing is forced at the first and the last move and (with a parity caveat) at the second moves; completing the induction is the concrete open problem.

12 Future work

1. Prove or refute Conjecture C1. A positive answer gives the exact count 2^{m-2} of resolutions of P_{2m} ; a negative answer would exhibit a resolution with a non-antipodal ear configuration, which our searches for $m \leq 7$ did not find.
2. Determine the maximum number of pairwise edge-disjoint p -angulations of P_n for $p \geq 4$ and n large; for quadrilaterals the exact values for $n \leq 10$ are 1, 3, 4, 5 for $n = 4, 6, 8, 10$, and the general answer is unknown to us.
3. Study the non-convex case: for a simple n -gon the number of triangulations varies wildly (from 1 to C_{n-2}), and the analogue of Theorem 1 should involve the number of *dissectible* diagonals rather than n .
4. Relate the resolution structure to cluster algebras: triangulations of a convex $2m$ -gon with the same word form an "interval" in a Tamari- or Cambrian-type lattice, and the m snakes of a resolution might correspond to a maximal antichain or to an almost-complete positive sequence in a rank-2 cluster algebra.

13 Conclusion

We determined the maximum number of pairwise edge-disjoint triangulations of a convex polygon to be $\lfloor n/2 \rfloor$, proved that for even n every extremal family is a partition of all diagonals into snakes with pairwise disjoint ear pairs, and classified the resolutions whose snakes follow a common word — these are automatically antipodal, and there are exactly 2^{m-2} of them for P_{2m} , in bijection with binary words of length $m-2$. The classification rests on a new and useful encoding of snakes as lattice paths, which also yields the count $n 2^{n-5}$ of snakes and makes the level structure of a snake visible. Exhaustive computation verifies that every resolution of P_{2m} follows a common word for $m \leq 7$ (polygons up to P_{14}); we conjecture the remaining case and locate precisely the step of the proof that fails.

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A Explicit resolutions of small polygons

For $n = 6$ the two resolutions (words LR and RL) are

$$\{(1, 3), (1, 4)\}, \{(0, 2), (2, 4)\}, \{(0, 3), (3, 5)\} \\ \{(1, 4), (2, 4)\}, \{(0, 2), (0, 3)\}, \{(1, 3), (1, 4)\}.$$

For $n = 8$ the four words are LLRR, LRLR, RLRL, RLLL, i.e. the first $m - 2 = 2$ letters are free. The library prints any of them:

```
>>> import edd
>>> res = edd.antipodal_resolutions(8)[1]      # word "LRLR"
>>> res.word, len(res)
('LRLR', 4)
>>> for T in res: print(sorted(T))
[(1, 6), (1, 7), (2, 5), (2, 6), (3, 5)]
[(0, 2), (2, 7), (3, 6), (3, 7), (4, 6)]
[(0, 3), (0, 4), (1, 3), (4, 7), (5, 7)]
[(0, 5), (0, 6), (1, 4), (1, 5), (2, 4)]
>>> edd.count_resolutions(8), edd.brute_force_count_resolutions(8)
(4, 4)
```

B Reproduction

```
pip install -e .
python -m pytest tests -q                # 188 tests
python falsification/run_falsification.py # 43 adversarial checks
python experiments/run_all.py            # regenerates results/ and figures/
```

The only numerical ingredient is HiGHS (through SCIPY); every other computation is exact integer arithmetic. The completeness check for $m = 8$ is available with `python experiments/run_all.py --full` but was not run for the tables in this repository, which stop at $m = 7$.